

CSE252 · INTRODUCTION TO ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

UNIT 1

Foundations of Artificial Intelligence

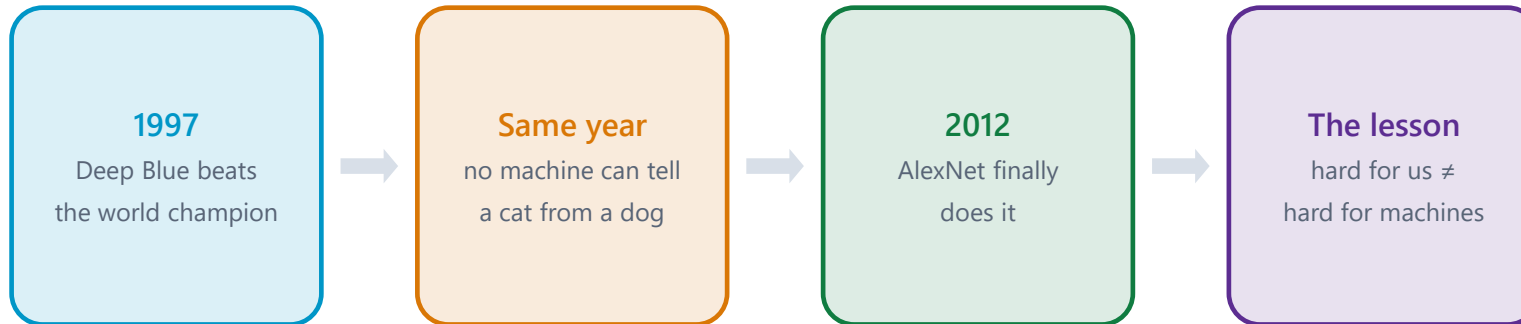
From Turing's question to Generative AI - what intelligence means for a machine, how we build it, and what we owe the people it affects.

Evolution of AI · Types of AI · AI vs ML vs DL vs DS · AI Lifecycle · Intelligent Agents · Applications · Ethics · Generative AI

OPENING STORY

The Machine That Beat the Champion - and the Four-Year-Old

UNIT 1

**HOW THE STORY GOES**

May 1997 IBM's Deep Blue defeats Garry Kasparov. Chess - the thing we treat as the summit of human reasoning - falls to a machine.

That same year no computer on earth could reliably look at a photograph and say whether it contained a cat or a dog.

A four-year-old does the second task instantly, and cannot play chess at all.

It took until 2012 for machines to do the four-year-old's task - fifteen years AFTER beating the champion.

So the ordering of 'hard' and 'easy' is completely different for us and for machines.

DID YOU KNOW?

This is called MORAVEC'S PARADOX: high-level reasoning needs surprisingly little computation, while low-level perception needs enormous amounts.

THINK OF IT LIKE a calculator that cannot recognise your face. brilliant at the thing you find hard, hopeless at the thing you find trivial.

TAKEAWAY What is hard for a human is often easy for a machine, and what a toddler does effortlessly took the field fifteen more years.

UNIT I ROADMAP

UNIT 1

What This Unit Covers - And Why It Comes First

Eight topics. Every later unit assumes the vocabulary you build here.

- Introduction to AI and its evolution
 - Why AI keeps being redefined - the 'AI effect'
- Types of Artificial Intelligence
 - Two independent axes: capability and functionality
- AI vs ML vs Deep Learning vs Data Science
 - Four terms, one nested hierarchy - not synonyms
- Problem-solving using AI and the AI Development Lifecycle
 - State-space formulation + the 6-phase build process
- Intelligent Agents and Agent Environments
 - PEAS specification and the 5 environment properties
- Applications: Robotics, Healthcare, Manufacturing, Smart Cities
- Ethics in AI and Responsible AI
- Introduction to Generative AI

WHAT YOU ALREADY BRING

What You Already Bring to This Unit

UNIT 1

Four bridges into this unit - including one from the course you are taking alongside this one.

FROM SCHOOL

Algorithms and logic

- You already write step-by-step procedures
- IF-THEN reasoning is the whole of expert systems
- AI's first thirty years were exactly this

FROM MATHEMATICS

Probability and functions

- A function maps input to output - that is a model
- Probability is how AI handles uncertainty
- You need no calculus for this unit

FROM DAILY LIFE

You already use AI

- Face unlock, Maps routing, spam filtering
- You have strong intuitions about what works
- Some of them are wrong - that is useful

WHERE IT LANDS

Your specialisation

- Perceive - decide - act IS a robot's control loop
- Section 5 builds the agent; section 6 flies it
- Every idea today has a robot to land on

THE FOUR BRIDGES

1 you can write an algorithm**2** you know what a function is**3** you use AI every day**4** and it all lands on a robot

LEARNING OUTCOMES

Learning Outcomes

UNIT 1

By the end of this unit you should be able to:

1

Define AI using the rational-agent view and name the four classical schools. **CO1**

2

Classify any system on both axes: capability and functionality. **CO1**

3

Distinguish AI, machine learning, deep learning and data science. **CO1**

4

Specify an agent with a PEAS description and characterise its environment. **CO1**

5

Explain algorithmic bias, the pillars of responsible AI, and what generative AI changes. **CO1**

Remember

recall the terms

Understand

explain them in your own words

Apply

use them on a problem you have not seen

Analyse

choose between them and justify it

the levels this unit is assessed at

01

SECTION

Introduction to AI and its Evolution

IN THIS SECTION

- What counts as 'intelligent' - and why the answer keeps moving
- The four schools: thinking/acting, humanly/rationally
- The Turing Test and its limits
- Seven eras from McCulloch-Pitts to Generative AI
- Why AI Winters happened - and what prevents the next one

1.1 INTRODUCTION TO AI

UNIT 1

Which of These Is 'Artificial Intelligence'?

Same question, four answers - and the honest answer changes every decade.

A**A thermostat**

- Reads temperature, switches the heater
- Follows one fixed rule
- Nobody calls this AI today

B**A chess engine**

- Searches millions of positions
- Beat the world champion in 1997
- Called AI in 1997 - 'just software' now

C**Face unlock**

- Learns your face from examples
- Works in the dark, with a beard, with glasses
- Was research-grade AI in 2014

D**ChatGPT**

- Writes essays, code and proofs
- Everyone calls this AI
- Ask again in 2035...

SHOW OF HANDS Vote A / B / C / D. Then defend your line: what exactly disqualifies the ones you rejected?

1.1 INTRODUCTION TO AI

Defining AI: The Rational Agent View

UNIT 1

KEY CONCEPT · The definition your exam actually wants

Artificial Intelligence the field of building systems that perceive their environment and take actions that maximise the chance of achieving their goals.

Intelligent agent anything that receives percepts from an environment and maps them to actions through an agent function $f : P^* \rightarrow A$.

Rational an agent is rational if, for every possible percept sequence, it selects the action expected to maximise its performance measure, given its built-in knowledge and what it has perceived so far.

Key move we do NOT define intelligence as 'human-like'. We define it as 'does the right thing', which is measurable and testable.

Read the notation carefully

- P^* = the set of ALL possible percept sequences (the entire history, not just the last input)
- A = the set of actions available to the agent
- Rationality depends on the performance measure - change the measure and the rational action changes

1.1 INTRODUCTION TO AI

Four Schools of AI - Two Questions, Four Answers

UNIT 1

Historically AI was defined along two axes: thinking vs acting, human vs rational.

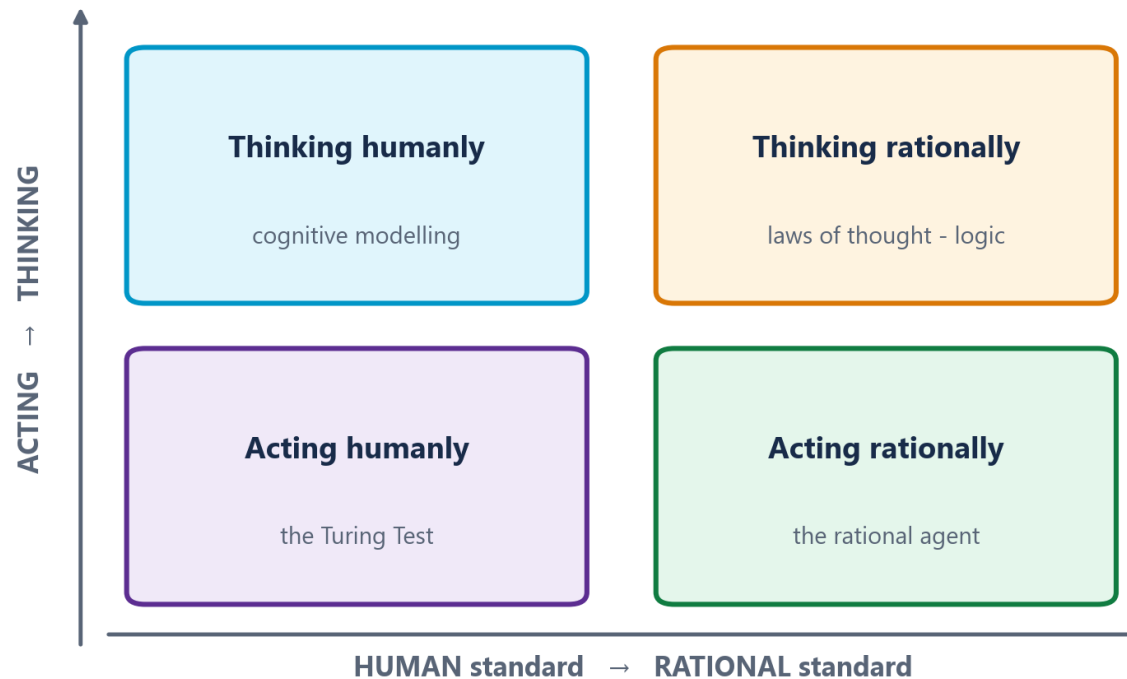
	ACT like a human	ACT rationally
THINK like a human	Cognitive modelling - replicate human reasoning steps (GPS, 1959)	Laws of thought - formal logic, provable inference
Measured by BEHAVIOUR	Turing Test - fool a human judge in conversation	Rational agent - do the right thing, whatever the mechanism

ASK THE CLASS Which quadrant does ChatGPT sit in? Which one does a self-driving car sit in? Are they the same?

1.1 INTRODUCTION TO AI

The Four Schools, on One Diagram

UNIT 1



HOW TO READ THIS FIGURE

Two axes thinking vs acting, and the human standard vs the rational standard.

Four boxes every historical definition of AI sits in one of them.

Bottom right is where modern AI lives - and where this course lives.

Why 'do the right thing' is measurable; 'be like a human' imports human error as the target.

TAKEAWAY AI is defined by ACTING RATIONALLY - the bottom-right box.

1.1 INTRODUCTION TO AI

The Turing Test - Brilliant, and Not Enough

UNIT 1

How the Imitation Game works

Setup a human interrogator holds a text conversation with two hidden parties - one human, one machine.

Pass condition the interrogator cannot reliably tell which is which.

Turing's point (1950) stop asking 'can machines think?' - it is undefinable. Ask a question you can actually run.

Capabilities implied natural language, knowledge representation, automated reasoning, and machine learning.

Total Turing Test adds vision and robotics so the machine must also perceive and manipulate objects.

Why modern AI moved past it

It rewards deception the winning strategy includes pretending to make typing errors and dodging hard questions.

It is not constructive passing tells you nothing about HOW to build a better system - no engineering signal.

Humans are a low bar in places and an impossibly high bar in others; human error is baked into the target.

ELIZA effect (1966) a 200-line pattern-matching script convinced users it was a therapist. Fooling people is easier than thinking.

Modern replacement task-specific benchmarks - ImageNet, GLUE, MMLU - which are measurable and comparable.

1.1 EVOLUTION OF AI

UNIT 1

The Evolution of AI - Part 1: Birth and First Winter

AI's history is not a smooth curve - it is boom, bust, boom, bust, boom.

1943-52**Gestational period**

McCulloch & Pitts model the first artificial neuron
Hebbian learning rule (1949)
Turing's paper and the Turing Test (1950)

1956**AI is named**

Dartmouth Workshop - McCarthy, Minsky, Rochester, Shannon
The term 'Artificial Intelligence' is coined
A 2-month, 10-man study... that ran for 60 years

1956-74**Golden era**

LISP language (McCarthy, 1958)
General Problem Solver (Newell & Simon)
ELIZA, SHRDLU, early search algorithms
Wild over-promising to funders

1974-80**First AI Winter**

Lighthill Report (UK, 1973) is damning
DARPA cuts funding
Causes: combinatorial explosion, no compute, machine translation failed

1.1 EVOLUTION OF AI

UNIT 1

The Evolution of AI - Part 2: Expert Systems to Generative AI

1980-87

Expert systems boom

MYCIN diagnoses blood infections
XCON saves DEC ~\$40M/year
Knowledge, not search: IF-THEN rules from experts
A billion-dollar industry

1987-93

Second AI Winter

LISP machine market collapses - cheap PCs won
Expert systems were brittle and could not learn
Knowledge acquisition bottleneck

1993-2011

Quiet, statistical AI

Deep Blue beats Kasparov (1997)
Focus shifts to probability and machine learning
The internet supplies data at last

2012-now

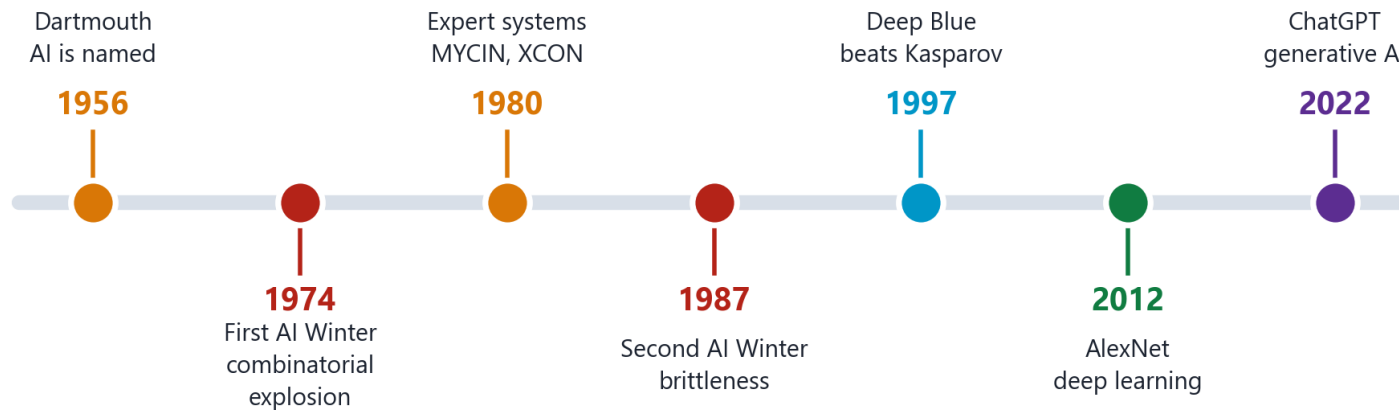
Deep learning & GenAI

AlexNet crushes ImageNet using GPUs (2012)
Transformers - 'Attention Is All You Need' (2017)
AlphaGo (2016), AlphaFold (2020)
ChatGPT, Gemini, Stable Diffusion

1.1 EVOLUTION OF AI

Seventy Years in One Line

UNIT 1



HOW TO READ THIS FIGURE

The shape, not the dates two booms, two winters, and the boom we are in now.

Red marks the winters 1974 and 1987 - and each had a specific technical cause.

Winter 1 combinatorial explosion. Good algorithms, impossible compute.

Winter 2 brittleness and the knowledge acquisition bottleneck.

2012 onwards data + GPUs + better training, all three at once.

TAKEAWAY AI's history is not a smooth curve - it is boom, bust, boom, bust, boom.

1.1 EVOLUTION OF AI

Why Did AI Winters Happen? (And Could One Happen Again?)

UNIT 1

COMMON PITFALL · The pattern repeats: over-promise, under-deliver, defund

The mechanism researchers over-promise to secure funding → hardware and data cannot deliver → funders lose confidence → money vanishes → progress stalls for a decade.

Winter 1 (1974-80) cause combinatorial explosion. The algorithms were sound but needed compute that did not exist.

Winter 2 (1987-93) cause brittleness and the knowledge acquisition bottleneck - hand-coded rules do not generalise and do not scale.

What is different now AI is commercially profitable at scale today. Winters ended when funding was research-grant-dependent; revenue is a sturdier foundation.

What could still trigger one unmet AGI hype, an energy/compute cost wall, a high-profile safety failure, or regulation after a public harm.

THINK-PAIR-SHARE 60 seconds with your neighbour: name one AI promise being made publicly today that you think will NOT be delivered by 2030.

1.1 EVOLUTION OF AI

UNIT 1

Case Study: Deep Blue vs Garry Kasparov, 1997

REAL-WORLD APPLICATION · The match that made the public believe

What happened IBM's Deep Blue defeated reigning world champion Garry Kasparov 3.5-2.5 in a six-game match - the first time a computer beat a world champion under tournament conditions.

How it won custom hardware evaluating ~200 million board positions per second, alpha-beta pruned search, and an evaluation function hand-tuned by grandmasters.

What it was NOT Deep Blue did not learn. It had no memory of past games and no understanding of chess. It was a Reactive Machine - brute-force search plus human expertise encoded by hand.

The 2017 contrast AlphaZero learned chess from nothing but the rules through self-play in 4 hours, and beat the best hand-crafted engine in the world.

The lesson Deep Blue is the peak of the 'encode human knowledge' era. AlphaZero is the 'learn it from data' era. That shift is this entire course.

CHECK YOUR UNDERSTANDING

Quick Check - Section 1

UNIT 1

QUICK CHECK

A system passes the Turing Test by convincing a human judge it is human. What does this prove about the system?

- A** It thinks rationally, using formal logic
- B** It can act in a way indistinguishable from a human in conversation
- C** It possesses genuine consciousness and understanding
- D** It has achieved Artificial General Intelligence

ANSWER: B The Turing Test measures ACTING HUMANLY - external behaviour only. ELIZA (1966) fooled users with simple pattern matching and understood nothing. Behaviour is not evidence of understanding, consciousness, or generality.

02

SECTION

Types of Artificial Intelligence

IN THIS SECTION

- Axis 1 - Capability: ANI, AGI, ASI
- Axis 2 - Functionality: Reactive, Limited Memory, Theory of Mind, Self-Aware
- Why every system you have ever used is Narrow AI
- Where the research frontier actually sits

1.2 TYPES OF AI

Classification by Capability - How Broad Is the Intelligence?

UNIT 1

Axis 1 asks: how WIDE is the competence?

AXIS 1 · LEVEL 1

Artificial Narrow Intelligence (ANI)

- Also called Weak AI
- Excellent at ONE task; useless outside it
- Cannot transfer learning to a new domain
- Examples: Siri, Google Translate, AlphaFold, Tesla lane-keeping, every chatbot, every recommender
- STATUS: 100% of AI in existence today

AXIS 1 · LEVEL 2

Artificial General Intelligence (AGI)

- Also called Strong AI
- Human-level ability across ANY domain
- Transfers knowledge between unrelated tasks
- Has common-sense reasoning and self-directed learning
- STATUS: hypothetical. No system is close.

AXIS 1 · LEVEL 3

Artificial Super Intelligence (ASI)

- Surpasses the best human minds in every field
- Better at science, art, strategy and social skill
- Raises the alignment and control problem
- STATUS: theoretical. Philosophy, not engineering.

POLL GPT-4 writes code, passes exams, analyses images and translates. Is it still Narrow AI? Justify your answer.

1.2 TYPES OF AI

Capability: Three Rungs, and We Are on the Bottom One

UNIT 1

ASI · Artificial Super Intelligence

surpasses the best humans in every field - theoretical

AGI · Artificial General Intelligence

human level in ANY domain - hypothetical, nothing is close

ANI · Artificial Narrow Intelligence

one task, superbly - 100% of all AI that exists

every real system you have ever used is on the bottom rung

HOW TO READ THIS FIGURE

Width, not depth this axis asks how BROAD the competence is - not how sophisticated.

ANI brilliant at one task, useless outside it. All of it. Everything.

AGI transfers knowledge between unrelated domains. Does not exist.

ASI beyond the best humans everywhere. Philosophy, not engineering.

The trap 'narrow' says nothing about how impressive a system is.

TAKEAWAY AlphaFold beats every human at protein folding and cannot play tic-tac-toe. That gap IS narrowness.

1.2 TYPES OF AI

Classification by Functionality - How Does It Handle Memory and Mind?

UNIT 1

Axis 2 asks: how DEEP is the machine's model of the world and of other minds?

Type	Core characteristic	Can it learn?	Real-world example
Reactive Machines	No memory of the past. Reacts only to the current input.	No	Deep Blue, spam filters, basic thermostats
Limited Memory	Stores recent history for a short window and uses it to decide.	Yes (trained)	Self-driving cars, chatbots, all modern ML
Theory of Mind	Understands that others have beliefs, emotions and intentions.	Research	Social/companion robots - active research, not solved
Self-Aware	Possesses consciousness and a model of its own internal state.	Theoretical	Does not exist. Philosophy, not engineering.

CLASSIFICATION DRILL

UNIT 1

Classify These Four Systems - Both Axes

THE PROBLEM

- A Tesla on Autopilot changing lanes on a highway
- IBM Deep Blue playing chess in 1997
- AlphaFold predicting a protein's 3-D structure
- The AI in a science-fiction film that fears being switched off

WHY IT MATTERS

- Exam questions almost always ask for BOTH axes - answer with both
- 'Narrow' describes breadth, not sophistication: AlphaFold is narrow AND world-class
- If a system learns from data, it is at least Limited Memory

STEP-BY-STEP SOLUTION

Tesla Capability = ANI (drives only). Functionality = LIMITED MEMORY - it tracks surrounding vehicles over the last few seconds to predict their paths.

Deep Blue Capability = ANI (chess only). Functionality = REACTIVE MACHINE - evaluates the current board with no memory of earlier games.

AlphaFold Capability = ANI (protein folding only - it cannot play chess). Functionality = LIMITED MEMORY - it learned from ~170,000 known structures.

Sci-fi AI Capability = AGI or ASI. Functionality = SELF-AWARE. Note this exists only in fiction - which is precisely why public intuition about AI is miscalibrated.

ANSWER Every real system above is ANI. Only the fictional one is not.

CHECK YOUR UNDERSTANDING

Quick Check - Section 2

UNIT 1

QUICK CHECK

A warehouse robot navigates aisles, remembers the last 30 seconds of obstacle positions, and cannot do anything except move stock. Classify it on BOTH axes.

A AGI + Limited Memory

B ANI + Reactive Machine

C ANI + Limited Memory

D ANI + Theory of Mind

ANSWER: C ANI because it does exactly one task and cannot transfer to any other. Limited Memory because it retains a short rolling history of obstacles to inform its next move - it is not purely reactive, but it builds no permanent world model and models no other minds.

03

SECTION

AI vs ML vs Deep Learning vs Data Science

IN THIS SECTION

- The nested hierarchy - and where Data Science overlaps rather than nests
- When learning from data beats writing rules
- Why deep learning needed 2012 and not 1986
- One fraud-detection system, four disciplines

1.3 AI VS ML VS DL VS DS

UNIT 1

The Nested Hierarchy - Four Terms, One Structure

AI $\supset$ ML $\supset$ DL. Data Science overlaps all three without sitting inside them.

OUTERMOST**Artificial Intelligence**

- Any technique that makes machines behave intelligently
- Includes rules, logic, search - not only learning
- A hand-written IF-THEN chess engine is AI with zero ML
- Born 1956

SUBSET OF AI**Machine Learning**

- Systems that learn patterns $y = f(x)$ from data
- No explicit rules written by a programmer
- Needs humans to engineer the features
- Units 3, 4 and 5 of this course

SUBSET OF ML**Deep Learning**

- Neural networks with many hidden layers
- Learns the features itself from raw pixels/audio/text
- Needs big data and GPUs
- Powers vision, speech and LLMs - Unit 6

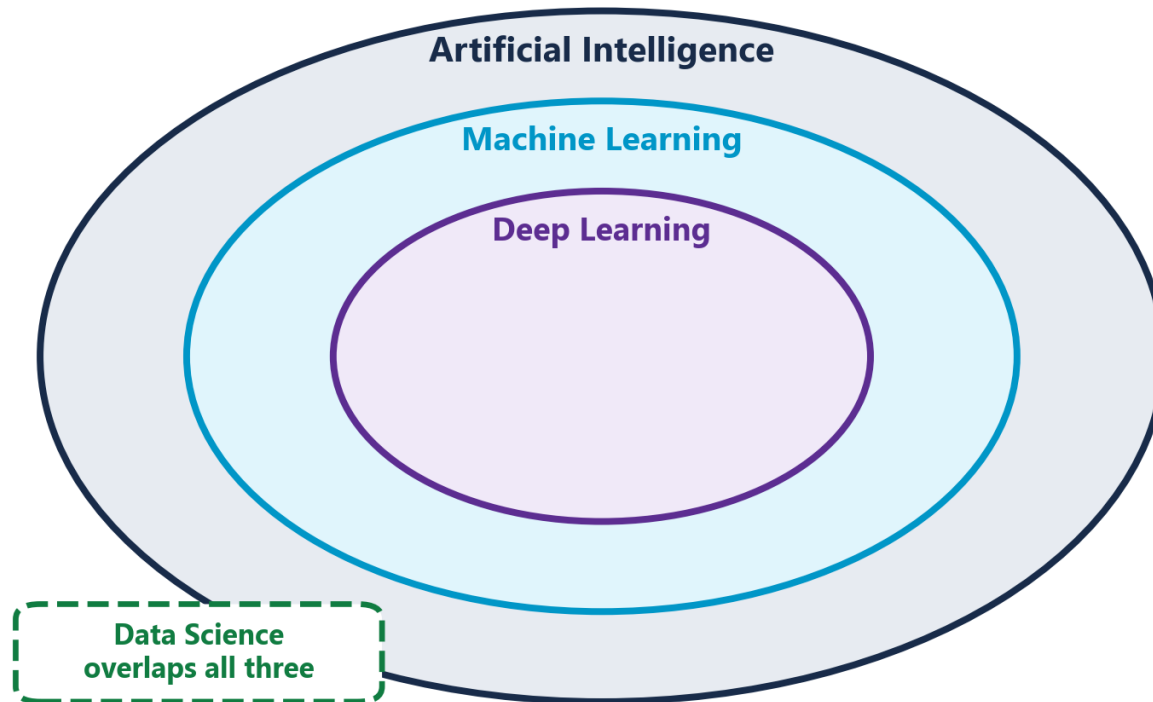
OVERLAPPING FIELD**Data Science**

- Extracting insight from data for decisions
- Statistics + domain expertise + engineering + ML
- Includes dashboards and A/B tests that use no AI
- NOT a strict subset - it overlaps AI

1.3 AI VS ML VS DL VS DS

The Nested Hierarchy, Drawn

UNIT 1



HOW TO READ THIS FIGURE

AI any technique that makes a machine behave intelligently - including rules and search.

ML $\subset$ AI learns patterns from data instead of being given rules.

DL $\subset$ ML neural networks that learn the FEATURES themselves.

Data Science the dashed box - it OVERLAPS all three rather than sitting inside them.

The proof Deep Blue is AI with zero machine learning. One example settles it.

TAKEAWAY Not all AI is machine learning. Deep Blue beat a world champion using search and hand-written rules.

1.3 AI VS ML VS DL VS DS

Traditional Programming vs Machine Learning

UNIT 1

The direction of the arrow is reversed - that is the entire idea of ML.

	Traditional Programming	Machine Learning
What YOU write	The rules, explicitly, by hand	Nothing - you supply examples instead
Inputs to the process	Data + Rules	Data + Answers (labels)
What comes out	Answers	Rules (the trained model)
Spam example	IF subject contains 'FREE' AND has 3+ links THEN spam	Show it 100,000 labelled emails; it derives the pattern itself
Fails when	Rules get too numerous or too subtle to write	Data is scarce, biased, or unrepresentative

TRY IT Write the rules to distinguish a cat from a dog in a photo. 60 seconds. Go.

1.3 AI VS ML VS DL VS DS

One System, Four Disciplines: Banking Fraud Detection

UNIT 1

REAL-WORLD APPLICATION · How AI, ML, DL and Data Science coexist in a single product

AI component a rule engine that automatically flags any transaction over \$10,000 originating overseas. Hand-written logic, no learning - and it catches real fraud on day one.

Machine Learning component a logistic regression model scoring the probability of fraud from the customer's historical spending pattern - amount, merchant category, time of day, distance from last transaction. (You will build this in Unit 3.)

Deep Learning component a Graph Neural Network tracing money-laundering rings across networks of shell companies - relationships no tabular model can see.

Data Science component the executive dashboard tracking fraud losses, false-positive rates and the cost of blocking legitimate customers. No model, pure insight.

The point these are not competing choices. A production system runs all four layers at once, each doing what it is best at.

DISCUSS The ML model flags your professor's card the moment they land abroad. Whose problem is that - and which layer should fix it?

CHECK YOUR UNDERSTANDING

Quick Check - Section 3

UNIT 1

QUICK CHECK

A team builds a hand-coded IF-THEN system that diagnoses equipment faults from sensor thresholds. It contains no learning of any kind. Is it AI?

- A** No - without machine learning it cannot be AI
- B** Yes - it is AI, specifically a rule-based expert system
- C** Only if the thresholds were derived statistically
- D** It is Data Science, not AI

ANSWER: B AI is any technique that makes a machine behave intelligently - including hand-written rules. This is a classic expert system, exactly the technology that drove the 1980s AI boom (MYCIN, XCON). ML is a SUBSET of AI, not a synonym for it.

04

SECTION

Problem-Solving using AI and the AI Development Lifecycle

IN THIS SECTION

- Formulating a problem as state-space search - the 5 components
- Worked formulation: the vacuum world and a grid robot
- The 6-phase AI Development Lifecycle (AIDLC)
- Why most AI projects fail at phase 1, not phase 4
- Python: encoding a problem as code

1.4 PROBLEM-SOLVING USING AI

UNIT 1

Formulating a Problem for AI: The Five Components

Before you can solve a problem with AI, you must define it formally. These five elements are that definition.

COMPONENT 1**Initial State S_0**

- Where the agent starts
- Robot at grid cell (0, 0)

COMPONENT 2**Actions $A(s)$**

- The legal moves available in state s
- {UP, DOWN, LEFT, RIGHT} minus any blocked by a wall

COMPONENT 3**Transition Model $\text{Result}(s, a)$**

- Which state results from doing action a in state s
- $\text{Result}((0,0), \text{RIGHT}) = (0,1)$

COMPONENT 4**Goal Test**

- A predicate: is this state a goal?
- Is the robot at the charging dock?

COMPONENT 5**Path Cost $c(s, a, s')$**

- The numeric cost of each step
- 1 per move, or metres travelled, or battery used

TOGETHER**= The State Space**

- All reachable states + the transitions between them
- A solution is a path from S_0 to a goal state
- An OPTIMAL solution is the lowest-cost such path

1.4 WORKED FORMULATION

UNIT 1

Worked Formulation: The Vacuum World

THE PROBLEM

- Two rooms, A and B
- Each room is either Clean or Dirty
- The vacuum sits in one room at a time
- Actions available: Left, Right, Suck
- Task: formalise this as a search problem

WHY IT MATTERS

- This is the canonical Russell & Norvig example - expect it in the exam
- Note the goal test is a PREDICATE, not a single state
- The state count $2 \times 2 \times 2$ comes from position $\times$ dirtA $\times$ dirtB

STEP-BY-STEP SOLUTION

States position $\times$ dirt status of both rooms = $2 \times 2 \times 2 = 8$ possible states. Example state: [vacuum in A, A dirty, B clean].

Initial state any of the 8 - the agent may start anywhere.

Actions A(s) {Left, Right, Suck} in every state.

Transition model Suck $\rightarrow$ the current room becomes Clean. Left $\rightarrow$ move to A (no effect if already in A). Right $\rightarrow$ move to B.

Goal test is every room Clean? Note: TWO of the 8 states satisfy this - vacuum in A with both clean, and vacuum in B with both clean.

Path cost 1 per action, so the cost of a path equals the number of actions taken.

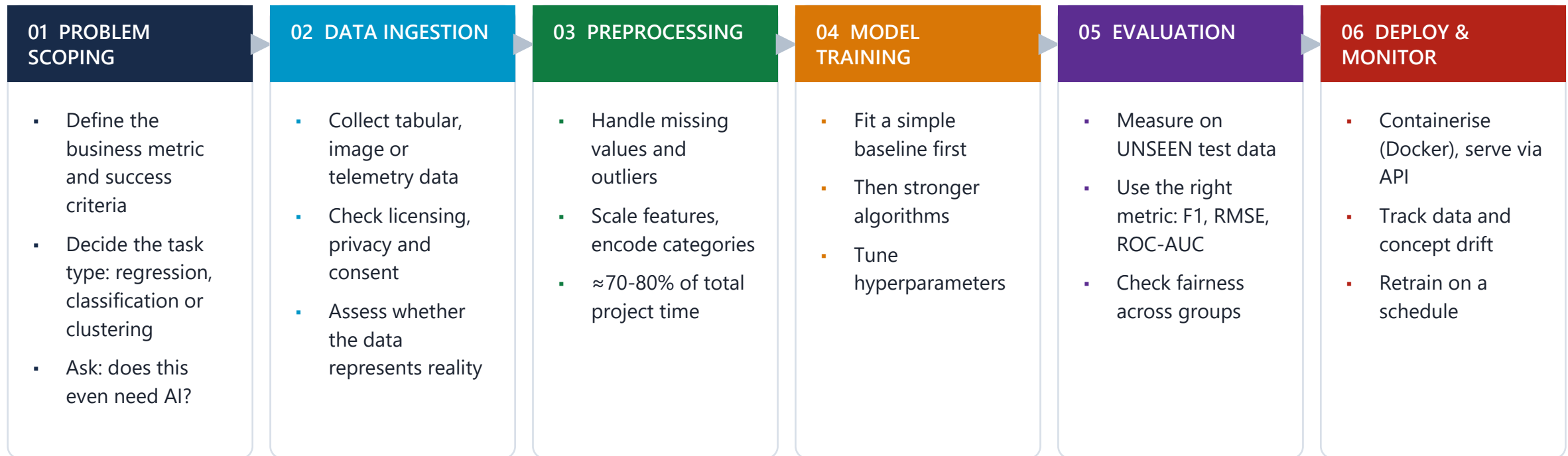
ANSWER 8 states, 2 goal states, cost = number of actions

1.4 AI DEVELOPMENT LIFECYCLE

UNIT 1

The AI Development Lifecycle (AIDLC) - Six Phases

It is a LOOP, not a line - monitoring in phase 6 sends you back to phase 1.

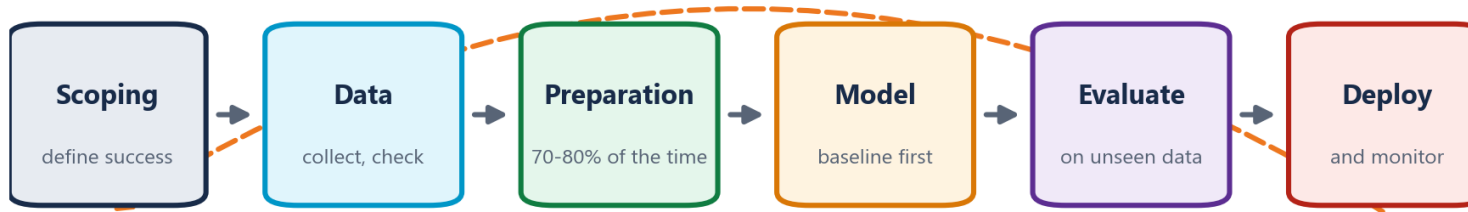


PREDICT Which phase consumes the most time in a real project? Vote before I reveal it.

1.4 AI DEVELOPMENT LIFECYCLE

The AI Lifecycle Is a Loop, Not a Line

UNIT 1



monitoring sends you back to scoping - the dashed arrow is the part people forget

HOW TO READ THIS FIGURE

Six phases left to right - but read the dashed arrow underneath.

Preparation is the fat one 70-80% of real project time. Nobody posts about it.

The dashed arrow deployment is not the end. Models decay as the world changes.

Data drift the inputs change. **CONCEPT DRIFT**: the input-outcome relationship changes.

Where projects die phase 1. Not phase 4. Solving the wrong problem beats every modelling error.

TAKEAWAY If nobody can say what decision changes when the system works, stop the project at phase 1.

Where AI Projects Actually Fail

COMMON PITFALL · Most failures happen before a single model is trained

Failure 1 - solving the wrong problem the model works perfectly and nobody uses it, because no decision in the business depends on its output.

Failure 2 - no success criterion 'improve customer experience' cannot be measured, so the project can never be declared finished or failed.

Failure 3 - unrepresentative data the training data does not match deployment reality. A model trained on one hospital's scanners fails at another hospital.

Failure 4 - data leakage information from the future or from the test set sneaks into training. Accuracy looks superb in development and collapses in production.

Failure 5 - no monitoring the model silently decays for months. Nobody notices until a business metric moves.

The uncomfortable truth these are process failures, not algorithm failures. Better maths would not have saved any of them.

1.4 AI DEVELOPMENT LIFECYCLE

UNIT 1

Activity: Scope an AI Project in 10 Minutes

GROUP WORK · 10 minutes

In groups of 3-4, take one problem and run it through phases 1-2 of the AIDLC. You are not building anything - you are proving you can define the problem.

1**Pick a problem**

- Predict which students are at risk of failing a course
- Detect defective items on a production line
- Route a delivery robot around a campus
- Or propose your own

2**Scope it (Phase 1)**

- What exactly is predicted?
- Regression, classification or clustering?
- What decision changes because of the output?
- How will you know it worked?

3**Plan the data (Phase 2)**

- What data do you need?
- Who owns it? Is it legal to use?
- What is likely missing or biased?
- Is there enough of it?

4**Find the trap**

- Name one way this project could fail
- Name one group it could treat unfairly
- Would a simple rule work instead of AI?

DELIVERABLE One group nominates a speaker: 60 seconds covering the task type, the success metric, and the biggest risk you identified.

05

SECTION

Intelligent Agents and the Agent Environment

IN THIS SECTION

- Agent = sensors + agent function + actuators
- The PEAS specification - and how to write one
- Five properties of task environments
- Why the environment dictates how complex your agent must be
- Types of agent: simple reflex to learning agents

1.5 INTELLIGENT AGENTS

UNIT 1

What Exactly Is an Agent?

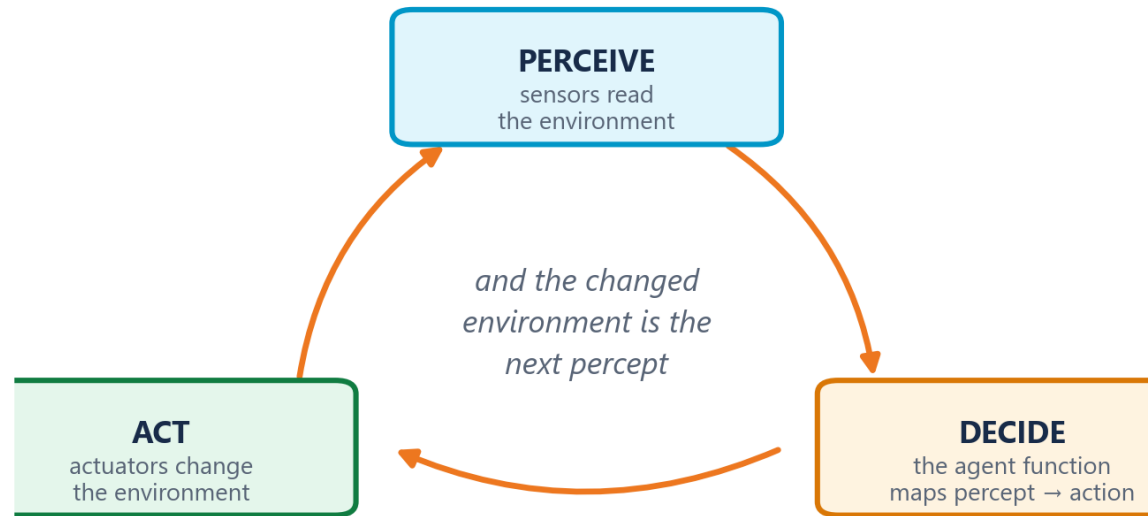
- An agent is anything that perceives its environment through SENSORS and acts upon that environment through ACTUATORS.
 - Human agent: eyes and ears are sensors; hands and legs are actuators
 - Robotic agent: cameras and LiDAR are sensors; motors and grippers are actuators
 - Software agent: keystrokes and network packets are sensors; screen output and file writes are actuators
- PERCEPT: what the agent senses at one instant. PERCEPT SEQUENCE: the entire history of everything it has ever perceived.
- AGENT FUNCTION $f : P^* \rightarrow A$ maps any percept sequence to an action. It is the abstract mathematical specification of behaviour.
- AGENT PROGRAM: the concrete code that implements the agent function on physical hardware.
 - The function is the specification; the program is the implementation
 - The function could be an infinite table; the program must be finite
- The agent's behaviour depends only on its percept sequence - it cannot act on what it has never perceived.

ASK THE CLASS Is a thermostat an agent? Is a calculator? Is a plant? What is the minimum requirement?

1.5 INTELLIGENT AGENTS

The Agent Loop - Perceive, Decide, Act

UNIT 1



HOW TO READ THIS FIGURE

Three steps, forever an agent does not run once and stop. It cycles.

PERCEIVE sensors. Camera, microphone, log file, keystroke.

DECIDE the AGENT FUNCTION - the mapping from percept to action. This is the part you build.

ACT actuators. And acting **CHANGES** the environment you will next perceive.

Why it is a loop that feedback is what separates an agent from a function.

TAKEAWAY Perceive → Decide → Act → the world has changed → perceive again. The loop is the definition.

1.5 INTELLIGENT AGENTS

The PEAS Framework - How to Specify an Agent

UNIT 1

KEY CONCEPT · Always specify the environment before you design the agent

P - Performance measure the quantitative criteria that define success. Chosen by the designer, NOT by the agent. Get this wrong and the agent optimises the wrong thing perfectly.

E - Environment the external world the agent operates in, including everything it must cope with - other traffic, weather, adversaries, noise.

A - Actuators the components through which the agent acts: steering, brakes, motors, a display, a database write, a message sent.

S - Sensors the instruments through which the agent perceives: cameras, LiDAR, GPS, keyboard input, log files, API responses.

Why it comes first PEAS is a design tool. You cannot choose sensors or algorithms until you have written down what success means and what the world contains.

The performance measure is the dangerous one

- Reward a vacuum for 'dust collected' → it dumps dust out and re-collects it
- Reward a delivery robot for 'speed' → it takes dangerous shortcuts
- Always measure what you actually want, not a convenient proxy for it

1.5 INTELLIGENT AGENTS

UNIT 1

PEAS Profiles for Three Real Agents

Read across each row - the four columns must be consistent with each other.

Agent	Performance measure	Environment	Actuators	Sensors
Autonomous car	Safety, legal trip, speed, comfort, fuel use	City roads, traffic, pedestrians, weather	Steering, throttle, brakes, indicators, horn	Cameras, LiDAR, radar, GPS, IMU, odometer
Medical diagnosis system	Diagnostic accuracy, patient outcome, cost, time to diagnose	Patient, symptoms, lab results, hospital staff	Display of diagnosis, test orders, referrals	Symptom input, patient history, lab and scan data
Warehouse sorting robot	Items sorted per hour, accuracy, zero collisions	Conveyor belts, storage bins, human workers, other robots	Robotic arm, gripper, drive wheels	Cameras, barcode scanner, pressure/force sensors, proximity
Interactive English tutor	Student's exam score improvement, engagement, retention	Students, curriculum, classroom or app	Screen output, exercise selection, corrections	Typed answers, response times, prior performance history

YOUR TURN Write the PEAS specification for a spam filter. Two minutes, then I take answers for each of the four letters.

1.5 AGENT ENVIRONMENTS

Properties of Task Environments - Five Questions to Ask

UNIT 1

PROPERTY 1

Fully vs Partially Observable

- Can the sensors see the complete state at any moment?
- FULLY: chess - the whole board is visible
- PARTIALLY: driving (blind spots, fog), medical diagnosis, poker
- Partial observability forces the agent to keep internal memory

PROPERTY 2

Deterministic vs Stochastic

- Is the next state fully determined by the current state and action?
- DETERMINISTIC: the vacuum world, an 8-puzzle
- STOCHASTIC: driving (a tyre may blow), robot wheels slip
- Stochastic environments need probability, not certainty

PROPERTY 3

Episodic vs Sequential

- Does the current decision affect future decisions?
- EPISODIC: spotting defects on a production line - each item is independent
- SEQUENTIAL: chess, driving - every move constrains the future
- Sequential environments require lookahead and planning

PROPERTY 4

Static vs Dynamic

- Does the world change while the agent is thinking?
- STATIC: a crossword puzzle waits for you
- DYNAMIC: taxi driving - traffic moves while you deliberate
- SEMI-DYNAMIC: chess with a clock - the world is fixed, your score decays

PROPERTY 5

Discrete vs Continuous

- Are states, time and actions countable or real-valued?
- DISCRETE: chess - finite squares, finite moves, distinct turns
- CONTINUOUS: steering angle, speed, camera input, elapsed time
- Continuous spaces cannot be enumerated - they must be approximated

ALSO WORTH KNOWING

Single vs Multi-Agent

- Are other agents present, and are they competing or cooperating?
- COMPETITIVE: chess, spam filtering (adversarial)
- COOPERATIVE: fleets of warehouse robots avoiding collisions
- Multi-agent environments require modelling other agents' intentions

1.5 AGENT ENVIRONMENTS

Chess vs Driving - Why One Was Solved and One Was Not

UNIT 1

Same five properties, opposite answers. The environment - not the algorithm - sets the difficulty.

	Chess	Autonomous driving
Observable?	FULLY - the entire board is visible	PARTIALLY - occlusions, fog, blind spots
Deterministic?	DETERMINISTIC - a move always has the same effect	STOCHASTIC - tyres slip, pedestrians are unpredictable
Episodic?	SEQUENTIAL - every move shapes the endgame	SEQUENTIAL - a lane change now constrains the exit later
Static?	SEMI-DYNAMIC - the board waits; the clock does not	DYNAMIC - the world moves while you compute
Discrete?	DISCRETE - 64 squares, finite legal moves	CONTINUOUS - steering angle, speed, position are real-valued
Solved?	YES - superhuman since 1997	NOT YET - after 25 years and billions spent

1.5 INTELLIGENT AGENTS

Types of Agent - Increasing Sophistication

UNIT 1

Five architectures, each solving a limitation of the one before it.

TYPE 1

Simple Reflex Agent

- Acts only on the CURRENT percept: condition-action rules
- IF dirty THEN suck
- No memory, no model of the world
- Fails badly in partially observable environments

TYPE 2

Model-Based Reflex Agent

- Keeps internal state to track what it cannot currently see
- Has a model of how the world evolves
- Handles partial observability
- Example: a robot remembering a room it already cleaned

TYPE 3

Goal-Based Agent

- Has explicit goals and searches for action sequences to reach them
- Considers the future, not just the present
- Flexible - change the goal, behaviour changes
- Example: satellite navigation planning a route

TYPE 4

Utility-Based Agent

- Goals say 'reach X'; utility says 'how GOOD is each way of reaching X'
- Trades off competing objectives - speed vs safety vs fuel
- Handles uncertainty by maximising EXPECTED utility
- Example: a car choosing between a fast risky route and a slow safe one

TYPE 5

Learning Agent

- Improves its own performance from experience
- Four parts: learning element, performance element, critic, problem generator
- The critic supplies feedback; the problem generator forces exploration
- This is what the rest of the course builds

THE PATTERN

Each type absorbs the last

- Reflex → adds memory → adds goals → adds preferences → adds learning
- Choose the SIMPLEST type your environment permits
- A thermostat does not need utility theory

CHECK YOUR UNDERSTANDING

Quick Check - Section 5

UNIT 1

QUICK CHECK

A robot vacuum cannot sense which room it is currently in, so it keeps an internal map of where it has already cleaned. Which agent type is it, and which environment property forced that design?

- A** Simple reflex agent; forced by a stochastic environment
- B** Model-based reflex agent; forced by partial observability
- C** Utility-based agent; forced by a continuous environment
- D** Learning agent; forced by a dynamic environment

ANSWER: B Keeping internal state to track what the sensors cannot currently see is the defining feature of a MODEL-BASED REFLEX AGENT, and the reason it is needed is PARTIAL OBSERVABILITY. A simple reflex agent, having no memory, would clean the same room forever.

06

SECTION

Applications of AI: Robotics, Healthcare, Manufacturing, Smart Cities

IN THIS SECTION

- Robotics and logistics - SLAM, AMRs, warehouse fleets
- Healthcare - imaging, drug discovery, AlphaFold, screening
- Manufacturing - visual inspection and predictive maintenance
- Smart cities - adaptive traffic, energy, waste
- What each domain has in common

1.6 APPLICATIONS - ROBOTICS

AI in Robotics and Logistics

UNIT 1

Robotics is where every part of this course meets the physical world.

CORE TECHNIQUE**SLAM**

- Simultaneous Localisation And Mapping
- Build a map of an unknown space WHILE tracking your position in it
- A chicken-and-egg problem: the map needs your position, your position needs the map
- Solved with probabilistic filtering (Kalman, particle filters)

DEPLOYMENT**Autonomous Mobile Robots**

- Amazon operates 750,000+ robots across its fulfilment centres
- Robots bring shelves to human pickers rather than the reverse
- Fleet coordination avoids gridlock among hundreds of units
- Multi-agent, dynamic, partially observable environment

DEPLOYMENT**Collaborative robots (cobots)**

- Work safely alongside humans without cages
- Force sensing stops the arm on contact
- Vision systems track human position continuously
- Safety is the dominant term in the performance measure

FRONTIER**Vision-Language-Action models**

- 'Pick up the red cup and put it in the sink' - in natural language
- One foundation model spanning perception, language and control
- Generalises to objects never seen in training
- The current research frontier in embodied AI

ROBOTICS LINK

The Robot Stack - Where Every Unit of This Course Lands

UNIT 1

SENSE cameras, lidar, IMU, encoders

raw, noisy, mismatched units → Unit II

PERCEIVE what is around me?

classify obstacles, read the scene → Units III and VI

MODEL what is normal here?

structure, maps, anomalies → Unit IV

DECIDE what should I do next?

agent, policy, plan → Units I and VI

ACT and MONITOR did that work?

measure, detect drift, improve → Unit V

*CO6 asks you to build one of these end to end***THINK OF IT LIKE** a driving lesson. eyes, judgement, decision, hands - and an instructor marking you.**TAKEAWAY** You are not learning six unrelated topics. You are learning one robot, one layer at a time.**HOW TO READ THIS DIAGRAM**This is **not** a metaphor these are the actual layers of a deployed robot.**Read the right-hand tags** each layer is a unit of CSE252.**Unit II** sensor data is the messiest data you will ever meet.**Unit V** on a robot, a wrong prediction moves something heavy.**CO6** your mini-project is assembling this stack.**DID YOU KNOW?**

A self-driving car runs all five of these layers about ten to sixty times per second, on every sensor, simultaneously.

1.6 APPLICATIONS - HEALTHCARE

UNIT 1

AI in Healthcare and Medicine

DIAGNOSTICS**Medical imaging**

- CNNs detect diabetic retinopathy from retinal photographs at specialist-level accuracy
- Deployed in India and Thailand where ophthalmologists are scarce
- Also: tumour detection in mammograms and CT, fracture detection in X-rays
- Positioned as decision SUPPORT - a clinician signs off

DRUG DISCOVERY**AlphaFold**

- Predicts a protein's 3-D structure from its amino-acid sequence
- Solved a 50-year grand challenge in biology
- 200M+ predicted structures released openly
- Cuts work that took years of lab crystallography to minutes

PREDICTION**Early warning systems**

- Predict sepsis hours before clinical symptoms appear
- Predict readmission risk at discharge
- Trained on vital signs, labs and notes in the patient record
- Every hour earlier in sepsis materially improves survival

PERSONALISATION**Precision medicine**

- Tailor treatment to a patient's genetic profile
- Predict which cancer therapy a specific tumour will respond to
- Reduces trial-and-error prescribing
- Raises hard questions about genomic data privacy

DISCUSS An AI screening tool is 95% accurate. Should a hospital deploy it without a doctor reviewing the results? What else do you need to know?

1.6 APPLICATIONS - MANUFACTURING & SMART CITIES

UNIT 1

AI in Manufacturing and Smart Cities

Two domains, one shared pattern: sensors generate data, models predict, systems act before failure.

MANUFACTURING

Automated Optical Inspection

- Computer vision inspects PCBs for solder defects and missing components
- Inspects thousands of units per hour, never tires, never blinks
- Detects flaws below the threshold of human vision
- Classic supervised image classification - Units 3 and 6

MANUFACTURING

Predictive maintenance

- Vibration, temperature and acoustic sensors on motors and turbines
- Predicts bearing failure weeks before it happens
- Replaces fixed-schedule servicing with condition-based servicing
- Unplanned downtime can cost more than \$250,000 per hour

SMART CITIES

Adaptive traffic control

- Cameras count vehicles per approach in real time
- Signal timing adapts to actual demand instead of a fixed cycle
- Reinforcement learning optimises across a whole junction network
- Documented reductions in waiting time and idling emissions

SMART CITIES

Energy, water and waste

- Forecast electricity demand and balance renewable supply
- Detect leaks from water-network pressure anomalies
- Route collection trucks only to bins whose sensors report them full
- Anomaly detection - an unsupervised task, Unit 4

1.6 APPLICATIONS

What Every Successful Application Has in Common

UNIT 1

KEY CONCEPT · The pattern behind all four domains

1. **A narrow, well-defined task** detect this defect, predict this failure, read this scan. None of these systems is general - all are ANI.
2. **Abundant, relevant data** millions of labelled images, years of sensor telemetry. No data, no model. This is why Unit 2 exists.
3. **A clear performance measure** defects caught, downtime avoided, waiting time reduced. If you cannot measure it, you cannot improve it.
4. **Tolerance for imperfection, or a human in the loop** the system is allowed to be wrong sometimes, either because errors are cheap or because a person reviews them.
5. **The cost of being wrong is understood** and the system is designed around that cost - which is a design decision, not a technical one.

APPLY IT Take the project your group scoped earlier. Does it satisfy all five? If one is missing, which one - and can you fix it?

07

SECTION

Ethics in AI and Responsible AI

IN THIS SECTION

- Where algorithmic bias actually comes from
- COMPAS and the automated hiring tool - two documented failures
- The pillars: fairness, accountability, transparency, privacy
- Explainability and the right to an explanation
- Privacy-preserving techniques and the engineer's responsibility

1.7 ETHICS IN AI

Algorithmic Bias: Where It Actually Comes From

UNIT 1

COMMON PITFALL · The model is not prejudiced - it is an accurate mirror of a biased past

The mechanism machine learning finds patterns in historical data and reproduces them. If the history encodes discrimination, the model learns discrimination as a rule - and applies it faster, cheaper and at greater scale than any human could.

Source 1 - Historical bias the data faithfully records a world that was already unfair. Past hiring favoured one group, so 'successful employee' correlates with that group.

Source 2 - Representation bias some groups are under-sampled. Facial recognition trained mostly on light-skinned faces performs far worse on dark-skinned faces - documented error-rate gaps of over 30 percentage points.

Source 3 - Measurement bias the proxy you measure is not the thing you care about. 'Arrests' is used as a proxy for 'crime', but arrests reflect policing patterns as much as criminal behaviour.

Source 4 - Deployment bias a model built for one purpose is used for another - a risk-assessment score designed to guide rehabilitation gets used to set prison sentences.

1.7 ETHICS IN AI

Two Documented Failures

UNIT 1

COMPAS - criminal risk scoring (USA)

What it did predicted a defendant's likelihood of reoffending; scores informed bail and sentencing decisions across several US states.

What investigation found a 2016 ProPublica analysis reported that Black defendants were roughly twice as likely to be incorrectly flagged high-risk, while white defendants were more often incorrectly flagged low-risk.

The technical twist the vendor showed the tool was calibrated - within each score band, reoffending rates were similar across groups.

Why both can be true researchers proved these fairness definitions are mathematically incompatible when base rates differ. You must choose which fairness you want - you cannot have all of them.

Lesson 'fair' is not one thing. It is a choice with consequences, and engineers must make it explicitly.

Automated resume screening

What it did an experimental tool at a major tech company ranked engineering applicants, trained on ten years of past hiring decisions.

What went wrong the industry's historical hiring skewed heavily male, so the model learned to downgrade resumes containing signals associated with women - including the word 'women's' as in 'women's chess club captain'.

The failed fix engineers removed explicit gender terms. The model simply relearned the pattern from PROXIES - certain sports, colleges, phrasing.

Why removal fails protected attributes are encoded redundantly across many correlated features. Deleting the label does not delete the signal.

Outcome the project was reportedly abandoned - which was the responsible decision. Sometimes the right answer is not to deploy.

THINK-PAIR-SHARE You are the engineer who discovers this bias two weeks before launch. What do you actually do - and who do you tell?

1.7 RESPONSIBLE AI

The Pillars of Responsible AI

UNIT 1

PILLAR 1

Fairness

- Predictions must not systematically disadvantage protected groups
- Measure error rates SEPARATELY per group - aggregate accuracy hides harm
- Choose your fairness definition explicitly and document why
- Tools: reweighting, resampling, fairness constraints, adversarial debiasing

PILLAR 2

Accountability & Governance

- A named human must be answerable for each automated decision
- Human-in-the-loop for high-stakes decisions - with real power to override
- Audit trails: what data, which model version, whose approval
- Model cards and datasheets documenting intended use and known limits

PILLAR 3

Transparency & Explainability

- Affected people deserve to know why a decision was made
- GDPR Article 22 constrains solely automated decisions with legal effect
- Techniques: LIME and SHAP (covered fully in Unit 5)
- Sometimes the right answer is a simpler, inherently interpretable model

PILLAR 4

Privacy & Data Protection

- Consent, purpose limitation, and data minimisation
- DIFFERENTIAL PRIVACY: add calibrated noise so no individual is identifiable
- FEDERATED LEARNING: train on-device; send model updates, never raw data
- Models can memorise and leak their training data

PILLAR 5

Safety & Robustness

- Behave predictably on inputs unlike the training data
- Adversarial examples: tiny pixel changes flip a classifier's answer
- Fail safely and visibly - degrade rather than collapse silently
- Test the edge cases, not just the average case

PILLAR 6

Sustainability

- Training large models consumes substantial energy and water
- Report the computational cost alongside the accuracy
- A smaller model that is 1% less accurate may be the better engineering choice
- Efficiency is an ethical property, not only a budget line

CHECK YOUR UNDERSTANDING

Quick Check - Section 7

UNIT 1

QUICK CHECK

A hiring model shows 94% overall accuracy. An audit finds it rejects qualified female candidates at three times the rate of equally qualified male candidates. Which statement is correct?

- A** The model is fine - 94% accuracy proves it works
- B** The model is broken and should be replaced with a rule-based system
- C** The model is accurate AND unfair; aggregate accuracy hides per-group harm
- D** Removing gender from the training data will solve the problem

ANSWER: C Accuracy and fairness are independent properties. A model can faithfully reproduce a biased history and be highly accurate while causing real harm. Aggregate metrics hide this - you must disaggregate by group. And (D) fails: the model relearns gender from proxies such as college, sports and phrasing.

08

SECTION

Introduction to Generative AI

IN THIS SECTION

- Discriminative vs generative models - what changes
- GANs and the min-max game
- Transformers, self-attention and Large Language Models
- Diffusion models for image synthesis
- Capabilities, limits, and where this is heading

1.8 GENERATIVE AI

Discriminative vs Generative Models

UNIT 1

Everything else in this course is discriminative. Generative models learn the distribution itself.

	Discriminative	Generative
What it learns	$P(Y X)$ - the boundary between classes	$P(X)$ or $P(X, Y)$ - how the data itself is distributed
The question it answers	'Given this image, is it a cat or a dog?'	'What does a cat look like? Draw me a new one.'
Output	A label or a number	A new sample that never existed
Examples	Logistic regression, SVM, random forest, most CNN classifiers	GANs, VAEs, diffusion models, Large Language Models
Where you meet it	Units 3 and 5 of this course	This section, and Unit 6

1.8 GENERATIVE AI

Generative Adversarial Networks - The Min-Max Game

UNIT 1

The idea that restarted image generation: make two networks compete.

THE GAN OBJECTIVE (GOODFELLOW ET AL., 2014)

$$\min_G \max_D V(D, G) = E[\log D(x)] + E[\log(1 - D(G(z)))]$$

Two networks trained against each other until neither can improve.

READING THE EQUATION

G the GENERATOR. Takes random noise z and produces a fake sample $G(z)$.

D the DISCRIMINATOR. Takes a sample and outputs the probability it is real.

x a real sample drawn from the true data distribution.

z random noise - the seed that makes each generated sample different.

min max G tries to MINIMISE what D tries to MAXIMISE. A two-player game.

At equilibrium D outputs 0.5 for everything - it can no longer tell real from fake. G has won.

1.8 GENERATIVE AI

Case Study: Synthetic Medical Images

UNIT 1

REAL-WORLD APPLICATION · Generative AI solving a data scarcity AND privacy problem at once

The problem training a diagnostic classifier for a rare disease requires thousands of labelled scans. For a rare condition, perhaps 200 exist worldwide - and every one is protected patient data that cannot leave its hospital.

The generative solution train a generative model on the available scans, then synthesise thousands of realistic new ones that belong to no real patient.

Why it works the classifier learns the visual signature of the disease from augmented data, without any real patient record being shared or exposed.

The privacy caveat generative models can MEMORISE and reproduce training examples. Synthetic does not automatically mean private - it must be tested for memorisation.

The clinical caveat a model trained mostly on synthetic data can learn the generator's artefacts rather than real pathology. Validation must use real, held-out patient data.

1.8 GENERATIVE AI

UNIT 1

Transformers and Large Language Models

One architecture from 2017 underlies essentially every system students call 'AI' today.

THE ARCHITECTURE**Transformer (2017)**

- 'Attention Is All You Need' - Vaswani et al.
- Replaced recurrent networks for sequence tasks
- Processes all tokens in parallel, not one at a time
- The foundation of GPT, Gemini, Claude, Llama

THE KEY IDEA**Self-attention**

- $\text{Attention}(Q, K, V) = \frac{\text{softmax}(QK^T / \sqrt{d_k}) \cdot V}{\text{softmax}(QK^T / \sqrt{d_k}) \cdot V}$
- Each word looks at every other word and weighs its relevance
- 'The animal didn't cross the street because IT was too tired'
- - attention resolves what 'it' refers to

HOW IT IS TRAINED**Next-token prediction**

- Objective: predict the next token, over and over, on internet-scale text
- No labels needed - the text labels itself (self-supervised)
- Then fine-tuned on instructions and human feedback (RLHF)
- Everything else - reasoning, code, translation - emerges from this

THE HONEST LIMITS**What LLMs cannot do**

- HALLUCINATION: fluent, confident, false statements
- No verified access to truth - they model plausibility, not fact
- Knowledge frozen at training cut-off unless given tools
- Struggle with exact arithmetic and long multi-step logic

ASK THE CLASS If an LLM is trained only to predict the next word, why can it write working code?

1.8 GENERATIVE AI

UNIT 1

Diffusion Models - How AI Image Generators Work

Learn to destroy an image, and you have learned how to create one.

STEP 1

The forward process

- Take a real image and add a little Gaussian noise
- Repeat hundreds of times
- End state: pure random static, no image left
- This part requires no learning at all

STEP 2

The reverse process

- Train a network to undo ONE step of noising
- Given a noisy image, predict the noise that was added
- Now you can run the whole process backwards
- Static → structure → recognisable image

STEP 3

Conditioning on text

- Guide the denoising with a text embedding
- 'An astronaut riding a horse' steers each step
- The text encoder and image model share a semantic space
- This is what makes prompting work

WHY IT BEAT GANS

Stability

- No adversarial game, so training is far more stable
- No mode collapse - it covers the full data distribution
- Higher fidelity and much better diversity
- Powers Stable Diffusion, DALL·E, Midjourney, Imagen

1.8 GENERATIVE AI

UNIT 1

Generative AI: What It Changes, and What to Watch

REMEMBER · A balanced view for engineers, not a hype cycle

Genuinely transformative code assistance, drafting and summarising, translation, design iteration, synthetic training data, accessibility tools, tutoring.

Genuinely limited no reliable access to truth, no persistent learning after training, weak at exact computation and long chains of logic, expensive to run.

The risks that matter hallucination presented fluently, deepfakes and misinformation, training-data consent and copyright, energy cost, over-reliance eroding skill.

Your professional stance use it as a fast, unreliable assistant. Verify everything that matters. Never let it be the last check on a decision that affects a person.

Where this course fits you cannot evaluate a generative system without the fundamentals - data quality (Unit 2), model behaviour (Units 3-4), and rigorous evaluation (Unit 5).

REFLECT Where in YOUR studies is generative AI genuinely useful - and where is using it actually costing you the learning?

COMMON MISTAKES

Common Mistakes to Avoid

UNIT 1

Every one of these is a real error students make in this unit - and every one is avoidable.

MISTAKE 1

Saying 'AI means machines that think like humans'

- Modern AI acts RATIONALLY, not humanly
- The four schools matter - name the right box

MISTAKE 2

Assuming all AI is machine learning

- AI $\supset$ ML. Deep Blue had zero learning
- Search and rules are AI too

MISTAKE 3

Giving one axis when asked to classify

- Capability AND functionality, every time
- 'ANI + Limited Memory', not just 'narrow'

MISTAKE 4

Treating a chatbot's fluency as understanding

- ELIZA fooled people with 200 lines
- Behaviour is not evidence of comprehension

THE FOUR THAT COST MARKS

X

'thinks like a human'

X

'all AI is ML'

X

one axis only

✓

check each before the exam

RESEARCH LENS

Research Lens

UNIT 1

Where this topic is heading - so you know what you are being prepared for.

DIRECTION 1**Agentic AI**

- Systems that plan and act over many steps
- Not one answer - a sequence of decisions
- The agent ideas in this unit, scaled up

DIRECTION 2**Multimodal models**

- One model handling text, images and audio together
- Closer to how perception actually works
- Blurs the neat categories you learned today

DIRECTION 3**The open question**

- Nobody agrees what would count as AGI
- Or how we would test for it
- The measurement problem is unsolved

WHERE THIS IS GOING

agents that plan over many steps



one model, many senses



and nobody agrees what AGI would mean

UNIT I RECAP

Unit I - The Eight Ideas Worth Remembering

UNIT 1

1

AI is defined by rational action, not human imitation

An agent perceives via sensors and acts to maximise a performance measure. 'Acting rationally' is measurable; 'thinking like a human' is not.

2

AI has boomed and busted twice - and the pattern is instructive

Winters followed over-promising. What ended them: data, compute and better algorithms arriving together around 2012.

3

Two independent axes classify every AI system

Capability (ANI / AGI / ASI) and Functionality (Reactive / Limited Memory / Theory of Mind / Self-Aware). All real systems today are ANI.

4

AI $\supset$ ML $\supset$ DL, with Data Science overlapping

ML learns rules from data instead of having them written. DL learns the features too. Not all AI is ML.

5

A problem must be formalised before it can be solved

Initial state, actions, transition model, goal test, path cost. Then the six-phase lifecycle - where 70-80% of the work is data.

6

PEAS specifies an agent; five properties specify its world

Observable, deterministic, episodic, static, discrete. The environment - not the algorithm - determines how hard the problem is.

7

Bias is an engineering failure mode, not a moral accident

Models faithfully reproduce historical injustice. Accuracy and fairness are independent - always disaggregate your metrics by group.

8

Generative models learn the distribution, not the boundary

GANs compete, Transformers attend, diffusion models denoise. All are powerful, all are fallible, none has access to truth.

NEXT UP Unit II - Data Preparation and the ML Workflow: NumPy, Pandas, Matplotlib, EDA, handling missing values and outliers, feature engineering, and train-test validation.